A FIVE-VERTEX COUNTEREXAMPLE TO CONJECTURE 5.8 OF CURTO, GENESON, AND MORRISON

Abstract. We refute the reverse implication of Conjecture 5.8 of Curto, Geneson, and Morrison concerning permitted motifs in composite CTLNs. We exhibit a five-vertex composite graph whose skeleton and all four components are permitted, whereas the full graph is forbidden, for every legal parameter choice. The construction is uniformly nondegenerate, and five vertices is the minimum possible order of such a counterexample.

1. The counterexample

For a directed graph G, the associated combinatorial threshold-linear network has connectivity matrix

$$W_{ij} = \begin{cases} 0, & i = j, \\ -1+\varepsilon, & j \to i, \\ -1-\delta, & j \not\to i, \end{cases} \qquad \theta > 0, \quad \delta > 0, \quad 0 < \varepsilon < \frac{\delta}{\delta+1}.$$

We follow the standing nondegeneracy convention of [1]. Thus the full vertex set is a permitted motif precisely when the solution of

$$(I - W_G)x = \theta\mathbf{1}$$

is strictly positive.

A composite graph is obtained by replacing each vertex of a skeleton by a component graph and inserting all intercomponent edges prescribed by the skeleton. Conjecture 5.8 of [1] states that if the skeleton is permitted, then the composite graph is permitted if and only if every component is permitted. The forward implication is Theorem 5.2 of [1]. Recent work gives a one-way compositionality theorem for low-rank gluings and a complete survival characterization for the rank-1 subclass [2]; the construction below is not rank 1, because vertices of G_B project differently to G_A and G_C.

Theorem 1. *There exists a composite graph G on five vertices such that, for every legal choice of $(\varepsilon, \delta, \theta)$, its skeleton and all four components are permitted motifs, whereas the full vertex set of G is forbidden. The CTLNs on G, its skeleton, and its components are nondegenerate throughout the legal parameter region. Consequently, the reverse implication in Conjecture 5.8 of* [1] *is false.*

Key words and phrases. combinatorial threshold-linear network, permitted motif, composite graph, counterexample.

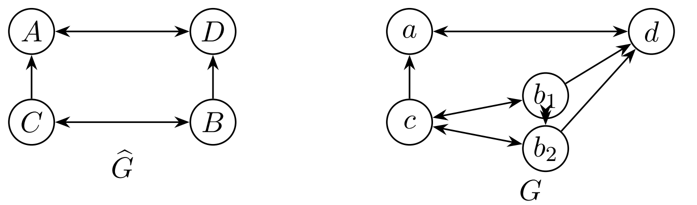

FIGURE 1. The permitted skeleton $\widehat{G}$ and its five-vertex composite graph G. The component G_B consists of b_1, b_2; the other components are singletons.

Proof. Let the skeleton $\widehat{G}$ have vertices A, B, C, D and edges

$$A \leftrightarrow D, \qquad B \leftrightarrow C, \qquad B \to D, \qquad C \to A.$$

Here K_m denotes the bidirected clique on m vertices. Take

$$G_A = K_1, \qquad G_B = K_2, \qquad G_C = K_1, \qquad G_D = K_1.$$

Writing the vertices of G as a, b_1, b_2, c, d, its edges are

$$a \leftrightarrow d, \qquad b_1 \leftrightarrow b_2, \qquad b_i \leftrightarrow c, \qquad b_i \to d \quad (i = 1, 2), \qquad c \to a.$$

Set

$$\alpha = 1 - \varepsilon, \qquad \beta = 1 + \delta.$$

In the orders (A, B, C, D) and (a, b_1, b_2, c, d), respectively,

$$I - W_{\widehat{G}} = \begin{pmatrix} 1 & \beta & \alpha & \alpha \\ \beta & 1 & \alpha & \beta \\ \beta & \alpha & 1 & \beta \\ \alpha & \alpha & \beta & 1 \end{pmatrix},$$

and

$$I - W_G = \begin{pmatrix} 1 & \beta & \beta & \alpha & \alpha \\ \beta & 1 & \alpha & \alpha & \beta \\ \beta & \alpha & 1 & \alpha & \beta \\ \beta & \alpha & \alpha & 1 & \beta \\ \alpha & \alpha & \alpha & \beta & 1 \end{pmatrix}.$$

Define

$$\widehat{h} = 2\delta^2 - 2\delta\varepsilon + 6\delta - \varepsilon^2 + 2\varepsilon,$$
$$h = 3\delta^2 - 3\delta\varepsilon + 9\delta - 2\varepsilon^2 + 4\varepsilon.$$

Direct multiplication verifies

$$(I - W_{\widehat{G}})^{-1}\theta\mathbf{1} = \frac{\theta}{\widehat{h}} \begin{pmatrix} \delta \\ 2\delta + \varepsilon \\ 2\delta + \varepsilon \\ \delta \end{pmatrix}, \qquad \det(I - W_{\widehat{G}}) = -\varepsilon^2 \widehat{h},$$

and

$$(I - W_G)^{-1}\theta\mathbf{1} = \frac{\theta}{\varepsilon h}\begin{pmatrix} -\delta^2 \\ \varepsilon(2\delta+\varepsilon) \\ \varepsilon(2\delta+\varepsilon) \\ \varepsilon(2\delta+\varepsilon) \\ \delta^2 + 3\delta\varepsilon + \varepsilon^2 \end{pmatrix}, \qquad \det(I - W_G) = -\varepsilon^3 h.$$

The legal inequalities imply $0 < \varepsilon < \min\{\delta, 1\}$, and hence

$$\widehat{h} = 2\delta(\delta - \varepsilon) + 6\delta + \varepsilon(2 - \varepsilon) > 0,$$

$$h = 3\delta(\delta - \varepsilon) + 9\delta + 2\varepsilon(2 - \varepsilon) > 0.$$

Thus $\widehat{G}$ is permitted, whereas the first coordinate of the full-support solution for G is negative. Hence G is forbidden.

The only nontrivial component is K_2, whose full-support solution is

$$\frac{\theta}{2-\varepsilon}(1,1) > 0.$$

Thus every component is permitted. Uniform nondegeneracy is proved in Appendix A. □

2. Minimality

The next reduction records the only information about a permitted component needed by the skeleton.

Lemma 2. *Let G be a composite graph with nonempty permitted components $G_1, \dots, G_N$, and suppose each $I - W_{G_i}$ is nonsingular. Put*

$$\alpha = 1 - \varepsilon, \qquad \beta = 1 + \delta,$$

and, for each i,

$$y_i = (I - W_{G_i})^{-1}\mathbf{1}, \qquad T_i = \mathbf{1}^\mathsf{T} y_i, \qquad d_i = T_i^{-1}.$$

Then

$$\alpha < d_i < \beta.$$

Let $A(d)$ be the $N \times N$ matrix with diagonal entries $A_{ii} = d_i$ and off-diagonal entries

$$A_{ij} = \begin{cases} \alpha, & j \to i \text{ in } \widehat{G}, \\ \beta, & j \not\to i \text{ in } \widehat{G}. \end{cases}$$

If $A(d)$ is nonsingular, then $I - W_G$ is nonsingular, and G is permitted if and only if $A(d)^{-1}\mathbf{1}$ is strictly positive.

Proof. The total-activity bound [1, Lemma 3.1], applied with input 1, gives

$$\beta^{-1} < T_i < \alpha^{-1},$$

so $\alpha < d_i < \beta$.

Skeleton edges	η	$\mathcal{Q}$
$\varnothing$	$+1$	$\{\delta+1,\ 2\delta-\varepsilon+3\}$
$1 \leftrightarrow 2$	-1	$\{\delta+1,\ \delta-\varepsilon+2,\ 2\delta-\varepsilon+3\}$
$1 \leftrightarrow 2,\ 1 \to 3$	-1	$\{1-\varepsilon,\ \delta+1,\ \delta-\varepsilon+2\}$
$1 \leftrightarrow 2,\ 1 \leftrightarrow 3$	-1	$\{1-\varepsilon,\ \delta-\varepsilon+2,\ \delta-2\varepsilon+3\}$
$1 \to 2 \to 3 \to 1$	$+1$	$\{\delta-\varepsilon+2,\ \delta-2\varepsilon+3,\ 2\delta-\varepsilon+3\}$
K_3	$+1$	$\{1-\varepsilon,\ \delta-2\varepsilon+3\}$

TABLE 1. Corner signs for the determinant and Cramer numerators of $A(d)$ for the six permitted three-vertex skeletons.

Let $S_i = \mathbf{1}^\mathsf{T} x_i$ be the total activity in component G_i. The block equations are

$$(I - W_{G_i})x_i = \left(\theta - \sum_{j \neq i} A_{ij} S_j\right)\mathbf{1}.$$

Writing the scalar in parentheses as r_i gives $x_i = r_i y_i$ and $S_i = r_i T_i$, hence

$$A(d)S = \theta\mathbf{1}.$$

Since $d_i > 0$ and $y_i > 0$, the block x_i is positive exactly when S_i is positive. This proves the positivity criterion.

For nonsingularity, let $(I - W_G)x = 0$ and define S_i as above. The same reduction gives $A(d)S = 0$, so $S = 0$. The block equations then give $(I - W_{G_i})x_i = 0$ for every i, whence $x = 0$. □

Theorem 3. *Every counterexample to Conjecture 5.8 of* [1] *has at least five vertices. Thus the construction in Theorem 1 has minimum possible order.*

Proof. We first rule out one, two, and three components. The case of one component is tautological. For two components, the only permitted skeletons are the independent set and the bidirected 2-clique [1, Figure 18]; both are covered by [1, Theorem 5.7].

It remains to exclude three components. Up to isomorphism, the six permitted three-vertex skeletons are those in the first column of Table 1; see [1, Figure 18]. Since $\theta > 0$ does not affect signs, set $\theta = 1$. For each skeleton, the determinant of $A(d)$ and its three Cramer numerators are multiaffine in (d_1, d_2, d_3).

At every corner $d_i \in \{\alpha, \beta\}$, direct substitution gives each Cramer numerator in

$$\{0, \eta(\delta+\varepsilon)^2\},$$

and the determinant in

$$\{0\} \cup \{\eta(\delta+\varepsilon)^2 q : q \in \mathcal{Q}\},$$

where η and $\mathcal{Q}$ are listed in Table 1. Each polynomial is nonzero at least one corner.

Every element of every displayed $\mathcal{Q}$ is positive in the legal parameter region. Multilinear interpolation expresses each value at an interior point of $(\alpha, \beta)^3$ as a convex combination of its eight corner values, with all weights positive. Therefore the determinant and all three Cramer numerators have the same strict sign η throughout that open box. Cramer's rule gives

$$A(d)^{-1}\mathbf{1} > 0.$$

By Lemma 2, no permitted three-vertex skeleton can produce a counterexample.

Thus a counterexample has at least four nonempty components. If it had at most four vertices, every component would be a singleton, so the composite graph would equal its permitted skeleton. Hence at least five vertices are necessary. $\square$

Appendix A. Uniform nondegeneracy

We verify the standing nondegeneracy hypothesis of [1, Definition 2.3]. Since $\theta > 0$ only multiplies each Cramer determinant by a positive factor, set $\theta = 1$.

Up to repetition over all principal submatrices of $I - W_G$, the distinct principal determinants are

$$\begin{gathered}1,\ -\delta(\delta+2),\ \varepsilon(2-\varepsilon),\ \varepsilon(\delta+1)-\delta,\ -\delta\varepsilon(\delta-\varepsilon+3),\\ \varepsilon^2(3-2\varepsilon),\ \varepsilon\big((2\delta+1)\varepsilon-2\delta\big),\ -\delta\varepsilon^2(\delta-2\varepsilon+4),\\ -\varepsilon(2\delta^2+4\delta+\varepsilon),\ -\varepsilon^2(2\delta^2-\delta\varepsilon+5\delta+\varepsilon),\ -\varepsilon^2\widehat{h},\ -\varepsilon^3 h.\end{gathered}$$

The distinct Cramer determinants are

$$\begin{gathered}1,\ \varepsilon,\ -\delta,\ \varepsilon^2,\ -\delta\varepsilon,\ -\delta\varepsilon^2,\ \delta^2\varepsilon^2,\ -\delta(\delta+2\varepsilon),\\ -\varepsilon(2\delta+\varepsilon),\ -\varepsilon^2(2\delta+\varepsilon),\ -\varepsilon^3(2\delta+\varepsilon),\ \delta^2+\delta\varepsilon+\varepsilon^2,\\ \varepsilon(2\delta^2+2\delta\varepsilon+\varepsilon^2),\ -\varepsilon(2\delta^2+4\delta\varepsilon+\varepsilon^2),\ -\varepsilon^2(\delta^2+3\delta\varepsilon+\varepsilon^2).\end{gathered}$$

The legal inequality implies

$$0<\varepsilon<1, \qquad \varepsilon<\delta, \qquad \varepsilon(\delta+1)-\delta<0, \qquad (2\delta+1)\varepsilon-2\delta<0,$$

where the last inequality follows from

$$\frac{\delta}{\delta+1} < \frac{2\delta}{2\delta+1}.$$

All remaining parenthesized factors are strictly positive, as are $\widehat{h}$ and h. Hence none of the displayed determinants vanishes. The corresponding determinant lists for $\widehat{G}$ are sublists of those above, while nondegeneracy of K_1 and K_2 is immediate.

References

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